

Tom S. Bertalan

RESEARCH SOFTWARE ENGINEER · BIOREACTOR CONTROLS · NEURAL NETWORKS FOR NONLINEAR DYNAMICS

Boston, MA

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*Machine learning and dynamical systems researcher specialized in neural system identification and unsupervised representations.
Proficient in handling high-dimensional time-series data for both offline parameter optimization and online execution.*

PROFESSIONAL EXPERIENCE AND RESEARCH AREAS

University of Massachusetts

RESEARCH SOFTWARE ENGINEER

Lowell, MA

1/2024-Present

• **Model Predictive Control of CHO Bioreactor**

- Design and administer virtualization, certificates, remote access, and OPC-UA communication for bioreactor distributed controls.
- Lead the rewrite of Java-based event-driven pump control logic, including regular review of junior engineers' work.
- Collect targeted experimental data from a 3-liter bioreactor, characterizing dynamics and responses to control inputs.
- Supervise the implementation of more rigorous simulation methods for a SBML-encoded bioreactor digital twin.
- Develop an optimal online model-predictive controller in Aspen DMC to maximize economic factors.

Johns Hopkins University

POSTDOCTORAL FELLOW

Baltimore, MD

2/2020-1/2024

• **Neural Differential Equations and Time-Series**

- Advanced SotA for neural DEs for time series, including CNNs for PDEs; conceived a novel loss function for Hamiltonian systems.
- Cut RNN inference burn-in from 25 to 5 samples using manifold learning.
- Derived custom gradients for nets with constraints; promoted theory on error scaling laws; applications in biomanufacturing.
- Parallelized neural PDE solution sensitivity analysis using SLURM on GPU cluster.

• **Modeling and Experimental Design for CHO Metabolism**

- Led a team of biophysics and ML experts in creating a suite of Python/MATLAB simulation and data processing tools.
- Created a GUI for Bayesian experimental design; mentored team members on its use and maintenance.

• **Robotic Perception and Control**

- Developed a variational autoencoder for end-to-end robotic localization.
- Trained a U-net on both open and custom datasets for real-time (>10hz) on-board semantic segmentation of drivable space.
- Inferred depth and semantic segmentation in simulation, using pre-trained transformer networks.
- Profiled depth/segmentation pipeline to ensure pure-GPU inference for real-time inference.

• **Machine Learning Service Deployment and Administration**

- Provisioned a local LLAMA2 inference server over VPN for diverse zero-shot microservices, including document renaming, topic extraction, and categorization.
- Automated conversion of seminar announcements to ICS files via OpenAI API, including curation of a 19.7k-word fine-tuning dataset.
- Administered lab GPU server for maximum uptime and ease of access for multiple concurrent users.
- Wrote specifications and solicited bids for an upcoming multi-GPU training server destined for datacenter colocation.

The Massachusetts Institute of Technology

POSTDOCTORAL ASSOCIATE

Cambridge, MA

3/2018-2/2020

• **Autonomous Vehicle Design and Pathfinding**

- Developed a model AV with firmware-level speed sensing and control commanded by checksummed bus communications.
- Designed a jerk-minimizing path planner capable of planing up to two lane switches ahead at 47 mph.
- Leveraged Ipopt and CppAD to push a model-predictive path follower to a latency of 67 ms in simulation.
- Taught summer courses on OpenCV, ROS, and CNNs for lane detection, path following, and traffic sign recognition.
- Wrote wrapper APIs for using video games as robotic simulations.
- Detected dashcam obstacles via windowed SVM.

• **Nonlinear dynamics in neuroscience**

- Wrote object-oriented library for fine- and coarse-grained simulation of neuronal dynamics.
- Analyzed bifurcation and resonance behavior of a mammalian circadian rhythm model.
- Played a key role in acquiring a \$1.8M grant for an industry-academic partnership.

Princeton University

NSF RESEARCH ASSISTANT

Princeton, NJ

9/2012-3/2018

Built a rover for particle-filter SLAM with LIDAR using Gazebo and ROS. Created a custom library for visualizing OpenCV pipelines and pruning of computational graphs at execution time. Simulated thousands of neurons in vectorized Numpy and MATLAB, and social agents in OpenMP-accelerated C++.

Created a 10-node Beowulf cluster in PHP, explored CUDA for PDE acceleration, and developed a LAMP-based social network. Simulated hierarchy formation in social animals as input to wet-lab experiments. Authored a multigrid subsurface-flow solver with CUDA acceleration.

SKILLS

Libraries, frameworks, & tools

- Python/C++/Java
- NumPy+SciPy+Matplotlib
- Pytorch/TensorFlow/Keras
- Linux, shell scripting, and git
- VSCode/Eclipse+PyDev
- Aspen DMC3 Controls
- OPC-UA, MODBUS/serial, and ROS

Areas of Expertise and Training

- Nonlinear dynamics of time series data
- Deep learning and ML
- Dimension reduction and representation learning
- Probabilistic modeling and theory
- Computer vision
- Research presentation and dissemination
- Trainee and peer mentoring

Other Skills and Interests

- Home automation with Arduino, Raspberry Pi, and 3D printing
- Solo and orchestral violin performance
- Windsurfing and small-boat sailing

EDUCATION & TRAINING

Institutions

- **Johns Hopkins University** **2020 - 2024**
Postdoctoral Fellow *Che. & Bio. Eng.*
- **The Massachusetts Institute of Technology** **2018 - 2020**
Postdoctoral Associate *Mech. Engr.*
- **Princeton University** **2012 - 2018**
NSF Research Fellow *PhD & MA, Che. & Bio. Eng.*
- **The University of Alabama** **2008 - 2012**
Student Research Assistant *BS Che. & Bio. Eng.; Minor in Math*

Awards

- Princeton Program in Plasma Science and Technology research and academic fellowship
- National Science Foundation research fellowship
- William R. Schowalter fund for scholarly conferences
- National merit finalist scholarship
- University honors program and president's list
- $\Phi\chi\Sigma$, $\Omega\chi E$, and $TB\Pi$ honor societies

Teaching

- Volunteered in a recurring summer course for high-school seniors and freshmen on employing ROS, OpenCV, and CNNs for lane detection and traffic sign recognition.
- Supervised operations and measurement activities in junior Chemical & Biological Engineering practicum.
- Lectured, held office hours, and graded assignments in senior differential equations course.

SELECTED PUBLICATIONS

Transformations establishing equivalence across neural networks: when have two networks learned the same task?

Tom Bertalan, F. Dietrich, I. Kevrekidis

2024
Chaos

Implementation and (Inverse Modified) Error Analysis for implicitly-templated ODE nets

A. Zhu, B. Zhu, Tom Bertalan, Y. Tang, I. Kevrekidis

2024
SIAMDS

Data-driven and Physics Informed Modelling of Chinese Hamster Ovary (CHO) Cell Bioreactors

T. Cui, Tom Bertalan, N. Ndauro, P. Khare, M. Betenbaugh, C. Maranas, I. Kevrekidis

2024
Comp. & Chem. Engr.

Certified Invertibility in Neural Networks via Mixed-Integer Programming

T. Cui, Tom Bertalan, G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab

2023
L4DC 2023 — PMLR

Learning emergent PDEs in a learned emergent space

F. Kemeth, Tom Bertalan, T. Thiem, S. Moon, C. Laing, I. Kevrekidis

2022
Nature Comm.

Initializing LSTM internal states via manifold learning

F. Kemeth, Tom Bertalan, N. Evangelou, T. Cui, S. Malani, I. Kevrekidis

2021
Chaos

Local conformal autoencoder for standardized data coordinates

E. Peterfreund, O. Lindenbaum, F. Dietrich, Tom Bertalan, M. Gavish, I. Kevrekidis, R. Coifman

2020
PNAS

On Learning Hamiltonian Systems from Data

Tom Bertalan, F. Dietrich, Igor Mezic, and I. Kevrekidis

2019
Chaos

PUBLICATIONS

Integration of Bayesian Optimization and Solution Thermodynamics to Optimize Media Design for Mammalian Biomanufacturing

Nelson Ndahiro, Edward Ma, [Tom Bertalan](#), Marc Donohue, Yannis Kevrekidis, and Michael Betenbaugh

In Press
iScience

Machine Learning Approaches to Problem Well-Posedness

[Tom Bertalan](#), G. Kevrekidis, E. Rebrova, S. Mishra, I. Kevrekidis

In Preparation

Data-driven and Physics Informed Modelling of Chinese Hamster Ovary (CHO) Cell Bioreactors

T. Cui, [Tom Bertalan](#), N. Ndahiro, P. Khare, M. Betenbaugh, C. Maranas, I. Kevrekidis

2024
Comp. & Chem. Engr.

Implementation and (Inverse Modified) Error Analysis for implicitly-templated ODE nets

A. Zhu, B. Zhu, [Tom Bertalan](#), Y. Tang, I. Kevrekidis

2024
SIAMDS

Transformations establishing equivalence across neural networks: when have two networks learned the same task?

[Tom Bertalan](#), F. Dietrich, I. Kevrekidis

2024
Chaos

Certified Invertibility in Neural Networks via Mixed-Integer Programming

T. Cui, [Tom Bertalan](#), G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab

2023
L4DC 2023 — PMLR

Some of the variables, some of the parameters, some of the times, with some things known: Identification with partial information

Saurabh Malani, [Tom Bertalan](#), T. Cui, M. Betenbaugh, J. Avalos, I. Kevrekidis

2023
Comp. & Chem. Engr.

Learning effective stochastic differential equations from microscopic simulations: linking stochastic numerics to deep learning

F. Dietrich, A. Makeev, G. Kevrekidis, N. Evangelou, [Tom Bertalan](#), S. Reich, I. Kevrekidis

2023
Chaos

Learning emergent PDEs in a learned emergent space

F. Kemeth, [Tom Bertalan](#), T. Thiem, S. Moon, C. Laing, I. Kevrekidis

2022
Nature Comm.

Personalized Algorithm Generation: A Case Study in Meta-Learning ODE Integrators

Y. Guo, F. Dietrich, [Tom Bertalan](#), D. Doncevic, M. Dahmen, I. Kevrekidis, Q. Li

2022
SIAM J. Sci. Comp.

Initializing LSTM internal states via manifold learning

F. Kemeth, [Tom Bertalan](#), N. Evangelou, T. Cui, S. Malani, I. Kevrekidis

2021
Chaos

Development of closures for coarse-scale modeling of multiphase and free surface flows using machine learning

C. Linares, [Tom Bertalan](#), E. Koronaki, Jicai Lu, G. Tryggvason, I. Kevrekidis

2021
APS Bulletin

Global and local reduced models for interacting, heterogeneous agents

T. Thiem, F. Kemeth, [Tom Bertalan](#), Carlo Liang, I. Kevrekidis

2021
Chaos

Local conformal autoencoder for standardized data coordinates

E. Peterfreund, O. Lindenbaum, F. Dietrich, [Tom Bertalan](#), M. Gavish, I. Kevrekidis, R. Coifman

2020
PNAS

Emergent spaces for coupled oscillators

T. Thiem, M. Kooshkbaghi, [Tom Bertalan](#), Carol Laing, I. Kevrekidis

2020
Front. in Comp. Neuro.

On Learning Hamiltonian Systems from Data

[Tom Bertalan](#), F. Dietrich, Igor Mezic, and I. Kevrekidis

2019
Chaos

An Emergent Space for Distributed Data with Hidden Internal Order through Manifold Learning

F. Kemeth, Sindre Haugland, F. Dietrich, [Tom Bertalan](#), Kevin Höhle, Q. Li, Erik Bollt, Ronen Talmon, Katharina Krischer, and I. Kevrekidis

2017
IEEE Access

Coarse-grained descriptions of dynamics for networks with both intrinsic and structural heterogeneities

[Tom Bertalan](#), W. Wu, C. Laing, C. William Gear, and I. Kevrekidis.

2017
Front. in Comp. Neuro.

Dimension reduction in heterogeneous neural networks: Generalized Polynomial Chaos (gPC) and ANalysis-Of-Variance (ANOVA)

M. Choi, [Tom Bertalan](#), C. Laing, and I. Kevrekidis.

2016
Euro. Phys. J., Special Topics

OpenMG: a new multigrid implementation in Python

[Tom Bertalan](#), A. Islam, R. Sidje, and E. Carlson

2014
Num. Lin. Alg. with App.

PRESENTATIONS

Symbolic regression and modular neural differential equations for bioprocess engineering and robotics

[Tom Bertalan](#), Jaeweon Lee, Zhao Wang, Seongkyu Yoon, I. Kevrekidis

2024
MLDS 4 (poster)

Certified Invertibility in Neural Networks via Mixed-Integer Programming

T. Cui, [Tom Bertalan](#), G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab

2023
Learning for Dyn. Sys.

Coarse-grained and emergent distributed-parameter systems from data

Hassan Arbabi, F. Kemeth, [Tom Bertalan](#), I. Kevrekidis

2021
American Control Conf.

Data-driven model reduction and discovery

T. Thiem, [Tom Bertalan](#), F. Kemeth, Yorgos Psarellis, I. Kevrekidis

2020
AIChE

Dynamical-systems-guided learning of PDEs from data

Hassan Arbabi, [Tom Bertalan](#), A. Roberts, I. Kevrekidis

2020
AIChE

On the data-driven discovery and calibration of closures

S. Lee, Yorgos Psarellis, C. Siettos, [Tom Bertalan](#), D. Amchin, T. Bhattacharjee, S. Datta, I. Kevrekidis

2020
AIChE

Connections between residual networks and explicit numerical integrators, and applications to identification of noninvertible dynamical systems

T. Cui, [Tom Bertalan](#), Yorgos Psarellis, I. Kevrekidis

2020
AIChE

Neural network approach to reduced order modeling of multiphase flows

C. Linares, [Tom Bertalan](#), S. Lee, J. Lu, G. Tryggvason, I. Kevrekidis

2020
APS Div. of Fluid Dyn.

PDE+PINN: Learning and Solving a PDE at the Same Time

[Tom Bertalan](#), F. Kemeth, T. Cui, I. Kevrekidis

2020
AIChE

Learning Partial Differential Equations from Discrete Space Time Data: Convolutional and Recurrent Networks, and Their Relations to Traditional Numerical Methods

[Tom Bertalan](#), F. Dietrich, T. Thiem, R. Farber, I. Kevrekidis, A. Roberts

2019
AIChE

Recurrent Neural Networks, Numerical Integrators and Nonlinear System Identification

[Tom Bertalan](#), R. Farber, T. Thiem, F. Dietrich, I. Kevrekidis

2018
AIChE

Coarse-Scale PDEs from Microscopic Observations Via Machine Learning

S. Lee, M. Kooshkbaghi, C. Siettos, [Tom Bertalan](#), and I. Kevrekidis

2019
AIChE

When Have Two Networks Learned the Same Task? Data-Driven Transformations between System Representations

[Tom Bertalan](#), F. Dietrich, T. Thiem, I. Kevrekidis

2019
AIChE

Coarse modeling of circadian rhythms in heterogeneous neural networks

[Tom Bertalan](#), C. William Gear, Yannis G. Kevrekidis, M. Henson, E. Herzog, and C. Laing

2017; 2016
Dyn. Days 2017; AIChE

Coarse-graining of heterogeneous neural dynamics

[Tom Bertalan](#), M. Choi, C. Laing, I. Kevrekidis

2015
AIChE

Heterogeneity and reduction for complex network dynamics

I. Kevrekidis, A. Holiday, [Tom Bertalan](#), and C. Laing

2014
AIChE

Coarse-graining Network Dynamics

A. Holiday and [Tom Bertalan](#)

2013
Network Front.

nSpyres: an open-source, Python-based framework for simulation of flow through porous media

E. Carlson, A. Islam, F. Dumkwu, and [Tom Bertalan](#)

2012
Interpore 2012

OpenMG: a new multigrid implementation in Python

[Tom Bertalan](#), A. Islam, R. Sidje, and E. Carlson

2012
Proc. 11th Python in Sci. Conf.

ESIM: a framework for simulation of dominance hierarchy formation in small animal groups

[Tom Bertalan](#) and R. Earley

2012
UA Hon. Undergr. Res. Conf.

An open-source computing cluster for virtual experiments with variable parameters

[Tom Bertalan](#) and E. Carlson

2011
UA Hon. Undergr. Res. Conf.